

Supporting Information

Structure–Activity Landscape Roughness Predicts Per-Compound QSAR Error Independently of the Applicability Domain

Krishna Harish

Elkins High School, Missouri City, Texas, United States
krishnaharish2009@gmail.com

This document contains: per-target Spearman correlations between each roughness descriptor and per-compound random-forest error, with the applicability-domain-controlled partial correlation and the activity-cliff fraction, for all 30 benchmark targets (Table S1); the full robustness sweep across neighborhood size k and molecular distance metric (Table S2); and extended graph-network architecture and training details with per-target errors (Table S3). All values are regenerated from the analysis pipeline by `src/make_si_tables.py`.

S1. Per-target correlations

For each target, the Spearman rank correlation ρ between each descriptor and the per-compound random-forest error is reported, computed on that target’s test set. “Cliff %” is the fraction of test compounds that participate in at least one benchmark-labeled activity cliff. “Nbr. disp. | AD” is the partial Spearman correlation of the activity-free neighborhood dispersion with error after controlling (by rank residualization) for nearest-neighbor similarity and local density. Dirichlet and Lipschitz use the query’s own activity; neighborhood dispersion and SALI density do not.

Table S1: Per-target Spearman correlations with per-compound random-forest error, and activity-cliff fractions, for all 30 MoleculeACE targets ($k = 10$, ECFP4/Tanimoto).

Target	n	Cliff %	Dirichlet	Lipschitz	Nbr. disp.	SALI mean	Nbr. disp. AD
CHEMBL1862_Ki	161	41.6	0.62	0.38	0.33	0.23	0.25
CHEMBL1871_Ki	134	23.9	0.76	0.51	0.32	0.21	0.25
CHEMBL2034_Ki	152	34.2	0.77	0.65	0.45	0.34	0.35
CHEMBL2047_EC50	128	39.1	0.69	0.51	0.34	0.28	0.29
CHEMBL204_Ki	553	39.8	0.64	0.53	0.40	0.30	0.41
CHEMBL2147_Ki	294	39.5	0.50	0.40	0.19	0.15	0.16
CHEMBL214_Ki	666	36.8	0.65	0.43	0.27	0.16	0.22
CHEMBL218_EC50	208	36.5	0.69	0.44	0.30	0.16	0.26
CHEMBL219_Ki	374	39.6	0.70	0.50	0.35	0.15	0.29
CHEMBL228_Ki	342	37.1	0.65	0.45	0.25	0.12	0.19
CHEMBL231_Ki	197	24.4	0.67	0.45	0.32	0.22	0.21
CHEMBL233_Ki	630	41.1	0.69	0.47	0.33	0.22	0.29
CHEMBL234_Ki	733	43.7	0.63	0.43	0.28	0.13	0.18

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Table S1 (continued)

Target	n	Cliff %	Dirichlet	Lipschitz	Nbr. disp.	SALI mean	Nbr. disp. AD
CHEMBL235_EC50	472	37.7	0.66	0.43	0.26	0.16	0.24
CHEMBL236_Ki	521	39.2	0.64	0.40	0.28	0.21	0.27
CHEMBL237_EC50	193	47.2	0.63	0.31	0.22	0.16	0.20
CHEMBL237_Ki	522	42.7	0.67	0.39	0.27	0.14	0.26
CHEMBL238_Ki	213	25.8	0.63	0.45	0.21	0.21	0.19
CHEMBL239_EC50	347	40.6	0.65	0.42	0.30	0.16	0.27
CHEMBL244_Ki	621	47.7	0.63	0.51	0.34	0.25	0.31
CHEMBL262_Ki	173	18.5	0.63	0.39	0.12	0.11	0.13
CHEMBL264_Ki	574	41.6	0.62	0.37	0.27	0.17	0.20
CHEMBL2835_Ki	126	10.3	0.74	0.59	0.49	0.36	0.20
CHEMBL287_Ki	267	38.6	0.61	0.38	0.19	0.01	0.15
CHEMBL2971_Ki	197	17.3	0.73	0.52	0.54	0.32	0.24
CHEMBL3979_EC50	226	41.6	0.60	0.41	0.10	0.02	0.05
CHEMBL4005_Ki	193	42.0	0.71	0.63	0.13	0.14	0.06
CHEMBL4203_Ki	149	8.7	0.62	0.61	0.19	0.11	0.16
CHEMBL4616_EC50	139	52.5	0.73	0.48	0.15	0.02	0.13
CHEMBL4792_Ki	297	53.9	0.72	0.47	0.23	0.16	0.24

S2. Robustness across neighborhood size and distance metric

The central correlations were recomputed for five neighborhood sizes k under three distance metrics: Tanimoto similarity over ECFP4 (radius 2) and ECFP6 (radius 3) fingerprints, and Euclidean distance in a standardized 12-descriptor physicochemical space. *local_var* is a uniform-weight landscape roughness (mean squared activity gap to the k neighbors, which uses the query’s activity); *nbr_disp* is the activity-free neighborhood dispersion; *nbr_disp | AD* is its partial correlation after controlling for nearest-neighbor distance and local density. Each cell is the median across the 30 targets. All fifteen k -metric combinations give a positive partial correlation.

S3. Graph-network details and per-target errors

Architecture. The graph model is a Graph Isomorphism Network (GIN-0). Each atom carries a 16-dimensional feature vector: a one-hot atomic-number indicator over {C, N, O, F, P, S, Cl, Br, I, other}, degree (scaled by 4), formal charge, an aromaticity flag, total hydrogen count (scaled by 4), and a ring-membership flag. Three message-passing layers of hidden width 64 update node states as $\mathbf{H}^{(l+1)} = \text{MLP}^{(l)}(\mathbf{H}^{(l)} + \mathbf{A} \mathbf{H}^{(l)})$, where $\mathbf{A}$ is the bond adjacency matrix and each $\text{MLP}^{(l)}$ is a two-layer perceptron with ReLU activation. The graph embedding is the sum over layers of the masked mean of node embeddings, read out by a two-layer perceptron head.

Training. Models are trained with Adam (learning rate 10^{-3}) to minimize mean squared error on training-standardized targets. The *fixed* regimen runs a flat schedule of 60 epochs. The *tuned* regimen holds out 10% of the training set as a validation split, trains for up to 150 epochs with learning-rate reduction on plateau and best-checkpoint early stopping (patience 20), and uses size-bucketed mini-batches of 128 to minimize padding. Per-compound test error is $|\hat{y}_i - y_i|$ in log-potency units. The tuned regimen lowers mean MAE on only 10 of 30 targets and does not change

Table S2: Median Spearman correlation (across 30 targets) with per-compound random-forest error, across neighborhood size and distance metric.

Metric	k	local_var	nbr_disp	nbr_disp AD
ECFP4 / Tanimoto	5	0.641	0.273	0.226
ECFP4 / Tanimoto	10	0.638	0.275	0.230
ECFP4 / Tanimoto	15	0.628	0.265	0.207
ECFP4 / Tanimoto	20	0.615	0.240	0.188
ECFP4 / Tanimoto	30	0.587	0.218	0.186
ECFP6 / Tanimoto	5	0.633	0.264	0.211
ECFP6 / Tanimoto	10	0.638	0.287	0.242
ECFP6 / Tanimoto	15	0.623	0.258	0.210
ECFP6 / Tanimoto	20	0.604	0.250	0.197
ECFP6 / Tanimoto	30	0.583	0.219	0.165
RDKit desc / Euclidean	5	0.464	0.149	0.135
RDKit desc / Euclidean	10	0.482	0.163	0.143
RDKit desc / Euclidean	15	0.457	0.134	0.111
RDKit desc / Euclidean	20	0.460	0.138	0.115
RDKit desc / Euclidean	30	0.461	0.131	0.104

the roughness ordering reported in the main text.

Table S3: Per-target mean absolute test error (log units) for the random forest and the two graph-network regimens, with the per-target ROGI global-roughness index.

Target	n	RF MAE	GNN (fixed-60)	GNN (tuned)	ROGI
CHEMBL1862_Ki	161	0.59	0.78	0.88	0.049
CHEMBL1871_Ki	134	0.53	0.61	0.71	0.056
CHEMBL2034_Ki	152	0.50	0.59	0.83	0.074
CHEMBL2047_EC50	128	0.42	0.64	0.69	0.078
CHEMBL204_Ki	553	0.55	0.80	0.74	0.058
CHEMBL2147_Ki	294	0.47	1.14	0.92	0.053
CHEMBL214_Ki	666	0.51	0.71	0.69	0.063
CHEMBL218_EC50	208	0.53	0.69	0.80	0.064
CHEMBL219_Ki	374	0.55	0.77	0.77	0.073
CHEMBL228_Ki	342	0.52	0.67	0.74	0.062
CHEMBL231_Ki	197	0.56	0.70	0.68	0.051
CHEMBL233_Ki	630	0.60	0.76	0.73	0.065
CHEMBL234_Ki	733	0.49	0.66	0.72	0.065
CHEMBL235_EC50	472	0.48	0.61	0.68	0.072
CHEMBL236_Ki	521	0.52	0.81	0.69	0.066
CHEMBL237_EC50	193	0.58	0.83	0.86	0.089
CHEMBL237_Ki	522	0.55	0.70	0.67	0.068
CHEMBL238_Ki	213	0.49	0.63	0.68	0.061

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Table S3 (continued)

Target	n	RF MAE	GNN (fixed-60)	GNN (tuned)	ROGI
CHEMBL239_EC50	347	0.51	0.68	0.94	0.085
CHEMBL244_Ki	621	0.56	0.80	0.76	0.069
CHEMBL262_Ki	173	0.57	0.75	0.76	0.063
CHEMBL264_Ki	574	0.48	0.62	0.66	0.066
CHEMBL2835_Ki	126	0.27	0.33	0.41	0.063
CHEMBL287_Ki	267	0.56	0.66	0.75	0.071
CHEMBL2971_Ki	197	0.43	0.58	0.67	0.043
CHEMBL3979_EC50	226	0.49	0.66	0.69	0.076
CHEMBL4005_Ki	193	0.50	0.66	0.87	0.072
CHEMBL4203_Ki	149	0.67	0.79	0.77	0.052
CHEMBL4616_EC50	139	0.51	0.65	0.76	0.110
CHEMBL4792_Ki	297	0.54	0.68	0.73	0.096

S4. Conformal prediction-interval coverage

90% inductive-conformal prediction intervals were built from the held-out per-compound residuals (main-text Methods). For each target the test compounds were split at random into equal calibration and evaluation halves. Coverage was made conditional on a per-compound variable in two standard ways: a locally-weighted (normalized) nonconformity score $|\hat{y}_i - y_i|/\sigma(x_i)$, and a Mondrian scheme that stratifies calibration into quintiles of the variable and calibrates within each. The conditioning variable was an activity-free roughness (the pairwise-SALI density) or the applicability domain ($1 - s_{\text{NN}}$). Every variant keeps the marginal coverage guarantee; they differ in where the coverage lands. Table S4 reports the median across the 30 targets over 50 random splits. Conditioning on roughness restores activity-cliff coverage toward the nominal level at a small width cost; conditioning on the applicability domain lowers it. The same pattern holds across the full roughness range, not only on the labeled-cliff subset: standard coverage falls monotonically from 0.95 in the smoothest roughness quintile to 0.87 in the roughest, roughness-Mondrian holds it within two points of nominal across all quintiles, and applicability-domain conditioning declines further (main-text Figure 6A; `results/conformal_curve.csv`).

Table S4: Median 90% conformal prediction-interval coverage and relative width across the 30 targets, by construction and conditioning variable.

Construction (conditioning variable)	Marginal	Cliff	Non-cliff	Median width (rel.)
Standard (unconditional)	0.904	0.872	0.920	1.00
Locally-weighted (roughness, SALI)	0.904	0.912	0.898	1.16
Locally-weighted (applicability domain)	0.904	0.820	0.954	1.14
Mondrian (roughness, SALI)	0.911	0.895	0.928	1.06
Mondrian (applicability domain)	0.910	0.861	0.945	1.01